

Cross-Task View Alignment for Multi-Task Continual Test-Time Adaptation

1. Motivation: diagnosing the failure modes of prior CTTA

Continual test-time adaptation (CTTA) has two flagship methods on the ACDC weather stream: **CoTTA** for segmentation (Wang *et al.* CVPR 2022) and **AMROD** for detection (Wei *et al.* ESWA 2026). Both use the same skeleton — student, EMA teacher, stochastic restore of student weights to source — and both produce a hallmark failure pattern under sustained continual drift: they perform reasonably on the first weather (fog) and **collapse on the last** (snow). CoTTA’s snow mIoU (35.79) is *below* its own source-only baseline (36.82). AMROD’s night AP50 (15.23) is barely +1.8 above source-only (13.45), while its later snow AP50 (40.99) recovers only because the domain happens to be visually closer to Cityscapes.

We argue that this collapse is not accidental — it is the structural consequence of using a **single head’s teacher-student consistency** as the sole supervision signal. Two specific failure modes:

FM-1. Teacher-collapse under a single supervision signal. In detection, teacher scores fall below AMROD’s threshold floor of 0.70 on night, no pseudo-boxes survive, and adaptation silently stops — the standard mean-teacher recipe has no orthogonal signal to notice the collapse. In segmentation, teacher per-pixel entropy grows and CoTTA’s soft-CE loss, weighting *every pixel equally*, amplifies teacher noise directly into gradient noise. Both are symptoms of the same problem: **when the teacher starts to drift, only the teacher’s own confidence can flag it, and it doesn’t.**

FM-2. Drift under uniform restore. CoTTA’s stochastic restore treats shared backbone and task-head weights identically. But the shared backbone is on the gradient path of every loss term, so it accumulates domain-specific bias faster than the restore reverts it. AMROD tries to fix this with a per-step Fisher-quantile restore (grad-squared as a proxy for parameter importance), but the per-step Fisher is extremely noisy — the same parameter can be “important” one step and “restored” the next. Both fail because they treat all shared-trunk drift the same way, whether the shared trunk actually drifted or only one head did.

The multi-task setting removes both failure modes at the source. When a shared-backbone model outputs both a detection map and a segmentation map, every object b is represented twice at the level of the shared backbone:

- as a **detection view** $z_{\text{det}}(b)$ (RoI-aligned FPN features), and
- as a **segmentation view** $z_{\text{seg}}(b)$ (mask-weighted pool of the same FPN features under the teacher’s per-pixel seg posterior for the segmentation class $\sigma(c)$ that corresponds to the box’s detection class c).

These two views are extracted by **different heads** from the **same shared representation**. If the shared representation is well-adapted, the two views agree up to a class-conditional embedding; if it is drifting, the two views diverge **before either head’s own confidence drops**. This disagreement is a target-domain-only self-supervised signal that no single-task method can access. It fires on FM-1 (either head’s collapse is visible in the other’s view before either becomes visibly wrong), and it enables a mechanistic response to FM-2 (shared-trunk drift is now distinguishable from head drift).

The paper builds a framework — **Cross-Task View Alignment (XVA)** — around this observation, with four primitives that jointly cover both failure modes:

Primitive	Failure mode addressed	Level	Section
CT-CL — SupCon between det and seg views	FM-1 (teacher-collapse, instance-level)	instance	§3
CT-CR — bbox interior → seg-class CE	FM-1 (teacher-collapse, dense-level)	pixel	§4
STR — shared-trunk stochastic restore split	FM-2 (drift under uniform restore)	weight	§5
CTPV — cross-task pseudo-label verification	FM-1 (teacher-collapse, pseudo-label-level)	pseudo-label	§5.5

All four are **defined only in the presence of two heads**. Applied to a single-task source model, CT-CL, CT-CR, and CTPV collapse to zero and STR degenerates to CoTTA’s uniform restore. The framework thus specialises gracefully — CTCMT_Det and CTCMT_Seg (§7) are what one gets when the framework runs on a single-task source, and they are still stronger than the corresponding published methods (Tables 1 and 2). We treat these as sanity checks; the paper’s central claim is the MTL result.

Fault modes FM-1 and FM-2 are formalised, along with code-level line references and numerical signatures, in §12.

1.1 Teacher-student plumbing (standard, briefly)

XVA is agnostic to the specific mean-teacher plumbing that generates the raw pseudo-labels; any teacher-student CTTA that provides per-image (`pseudo_boxes`, `teacher_seg_probs`) will do. We use a standard CoTTA-with-AMROD-detector setup for the reported experiments, described compactly below and reusable across all XVA variants:

- **Student θ_s** : trainable copy of the source Panoptic-FPN.
- **Teacher θ_t** : EMA of student, $\theta_t \leftarrow \beta \theta_t + (1 - \beta) \theta_s$, $\beta = 0.9998$.
- **Anchor θ_0** : frozen copy of source weights, source of the sparse restore.
- **Detection pseudo-labels**: per-class dynamic thresholds $\tau_c^{(t+1)} = \gamma \tau_c^{(t)} + (1 - \gamma) \alpha \sqrt{\bar{s}_c^{(t)}}$ as in AMROD; kept boxes become `gt_instances` for standard Faster R-CNN losses.
- **Segmentation pseudo-labels**: raw teacher softmax; standard soft-CE.
- **Score-EM gate**: skip the step when the teacher-score EMA is stable (image already adapted).

None of the above is our contribution. **Everything in §2 onward is.**

2. Cross-task view construction

For every pseudo-detected box b of class c , we construct two feature vectors from the *student* FPN feature maps F :

Detection view

$$z_{\text{det}}(b) = \phi_{\text{det}}\left(\text{RoIAlign}(F, b)\right)$$

where ϕ_{det} is a lightweight L2-normalised projection head (single Conv + GELU + Linear, 128-d output).

Segmentation view (novel construction)

Let $p_{i,\sigma(c)}^{\text{tch}}$ be the teacher’s segmentation posterior at pixel i for the segmentation class $\sigma(c)$ that corresponds to detection class c under the Cityscapes taxonomy (`_DET_TO_SEG_CLASS_CITYSCAPES`, e.g. `det-person` $\rightarrow$ `seg-person`, `det-car` $\rightarrow$ `seg-car`, ...). The segmentation view is the teacher-posterior-weighted mean of student FPN features **inside the box only**:

$$z_{\text{seg}}(b) = \phi_{\text{seg}} \left(\frac{\sum_{i \in b} p_{i,\sigma(c)}^{\text{tch}} F_i}{\sum_{i \in b} p_{i,\sigma(c)}^{\text{tch}} + \varepsilon} \right)$$

with ϕ_{seg} another 128-d projection head. The crucial detail: z_{seg} is **weighted by the seg posterior but the features F_i come from the student**, so gradients flow into the shared backbone through both views. The teacher-driven mask concentrates the pool on genuinely class-relevant pixels, so the view is robust to bbox looseness.

Why this specific construction is novel. Prior cross-task learning (e.g. Cross-Stitch, MTL-NAS, PAD-Net) shares features across tasks *during source training*. Prior CTTA work uses at most one head per image. Ours is the first to (a) *build two views per-object at inference time*, and (b) *use their agreement as the adaptation signal*, not as an auxiliary supervision for source training.

3. Instance-level XVA — CT-CL

Given all boxes in the current image as $\{(z_i, y_i)\}$ with $z_i \in \{z_{\text{det}}, z_{\text{seg}}\}$ and y_i the detection class of the underlying box, we apply supervised contrastive loss (Khosla *et al.* 2020) over the combined view pool:

$$\mathcal{L}_{\text{CT-CL}} = - \sum_i \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \neq i} \exp(z_i \cdot z_a / \tau)}$$

with $\tau = 0.07$. The positive set $P(i)$ contains three qualitatively different pair types:

Pair type	Effect on the shared representation
$z_{\text{det}} \leftrightarrow z_{\text{det}}$ (same class)	classical detection SupCon
$z_{\text{seg}} \leftrightarrow z_{\text{seg}}$ (same class)	classical segmentation SupCon
$z_{\text{det}} \leftrightarrow z_{\text{seg}}$ (same class)	cross-task pull — the XVA signal

The third row is what forces the two heads’ post-projection embeddings to converge on a **shared per-class code**. It is the loss term without which XVA reduces to two independent SupCon losses that could equivalently be optimised in single-task pipelines.

4. Dense XVA — CT-CR

CT-CL is instance-level: it pulls average box-embeddings together but says nothing about the *pixels* inside each box. CT-CR closes that gap with a dense supervision that costs one extra

loss term and no new parameters. For every pseudo-box b of class c , all pixels inside b should be labelled by the student segmentation head as $\sigma(c)$:

$$\mathcal{L}_{\text{CT-CR}} = \sum_{i \in b} \text{CE}(p_i^{\text{stu-seg}}, \sigma(c))$$

Pixels outside every box are ignored (label 255). This propagates detection-side certainty back through the seg head and, symmetrically, enforces that the seg head’s per-pixel predictions be spatially consistent with the detection geometry.

Instance $\times$ dense granularity is the point. CT-CL and CT-CR target different failure modes: - CT-CL prevents the shared representation from becoming *class-degenerate* across tasks (each head could still classify well but from different features). - CT-CR prevents the seg head’s output from becoming *spatially incoherent* with detected objects (each object could still be detected but with wrong pixel support).

Their ablation deltas are additive in the same direction on both metrics (Table 4, §11), which is the empirical signature of two mechanisms targeting independent problems.

5. Weight-level XVA — Shared-Trunk Restore (STR)

CoTTA’s stochastic restore replaces a random fraction ρ of student weights with their anchor copy after each step. In a **single-task** model this rate is well-calibrated: one loss produces one gradient flow through the whole model, and a flat restore rate returns each weight to its source at a rate proportional to how often it is updated.

In an MTL model this calibration breaks. The **shared trunk** (backbone + FPN) receives gradients from det and seg losses **plus** CT-CL and CT-CR; the **task-specific heads** each receive gradients from only their own loss plus (partially) CT-CR. A flat restore rate therefore under-corrects the trunk relative to the heads. STR corrects this proportionally:

$$\text{Pr}[\text{restore } \theta_i] = \begin{cases} \rho \cdot \eta & \text{if } \theta_i \in \text{backbone} \cup \text{FPN (shared)} \\ \rho & \text{otherwise (task heads)} \end{cases}$$

with $\rho = 0.01$ and $\eta = 0.1$. Note the direction of the correction: $\eta < 1$ means the shared trunk gets restored **more often** than the heads per weight (because gradient traffic is heavier), preventing the shared representation from silently accumulating multi-task drift.

STR is one-parameter (η), no extra memory, and the shared-trunk distinction is available from the model definition — no learning required. In the code it is called V2 (task-aware stochastic restore) and gated by CTCMT_CROSS_TASK_FISHER; we call it STR in the paper to emphasise the principle over the ordinal name.

Why it’s not just a flat hyperparameter re-tune. A universal restore rate $\rho' < \rho$ would slow *everything* down and hurt heads. Task-specific restore requires the shared/task-head distinction that MTL provides. Empirically STR alone contributes **+0.80 AP50 / +0.62 mIoU** over the flat-CoTTA baseline on ACDC continual (Table 4).

5.5 Pseudo-label-level XVA — Cross-Task Pseudo-label Verification (CTPV)

CT-CL and CT-CR operate on the *pseudo-boxes emitted by the teacher* as if those boxes were ground truth. But the teacher’s detection head can be wrong about the *class* of a box even when its bounding coordinates are accurate — a “bicycle” detected on a motorcycle image, a “car” on a partially-occluded truck. In those cases every downstream XVA loss inherits a false

class label: CT-CL pulls incorrect positives together, CT-CR supervises seg pixels toward the wrong class, and the shared trunk drifts toward the wrong equilibrium.

The MTL setting exposes an independent second opinion on the class of every detected box: the *segmentation head’s own posterior on the pixels inside that box*. CTPV consults it. For each pseudo-box b of class c we compute the seg head’s mean posterior on class $\sigma(c)$ over the box interior:

$$q(b, c) = \frac{1}{|\Omega_b|} \sum_{i \in \Omega_b} p_{i, \sigma(c)}^{\text{tch-seg}}$$

and reject the box if $q(b, c) < \tau_{\text{CTPV}}$ (default $\tau_{\text{CTPV}} = 0.3$). Surviving boxes flow into CT-CL, CT-CR, and the detection loss unchanged; rejected boxes are removed from the batch.

CTPV is a *pseudo-label-level* XVA primitive, i.e. it acts before any loss is computed, filtering the supervision signal by cross-task agreement. It is the fourth granularity of XVA:

Level	Signal	Enforcement
Instance (CT-CL)	$z_{\text{det}} \leftrightarrow z_{\text{seg}}$	attractive/repulsive contrastive
Dense (CT-CR)	pixel-in-bbox $\rightarrow$ seg-class	dense cross-entropy
Weight (STR)	shared trunk vs task heads	asymmetric restore rate
Pseudo-label (CTPV)	teacher det $\leftrightarrow$ teacher seg	veto

Why it works and where. CTPV fires exactly when a teacher’s det and seg heads disagree about an object’s class. On the fog domain (where both heads are near-source-optimal) it rejects <2% of boxes. On the snow domain (last in the continual sequence, where accumulated drift is largest) it rejects ~7% of boxes — precisely the images where teacher-collapse (FM-1) starts to leak wrong labels into XVA. Empirically CTPV alone contributes **+0.23 AP50 / +0.28 mIoU** over the STR+CT-CL+CT-CR baseline on ACDC continual, with the largest per-domain gains on snow (+0.65 AP50 / +0.84 mIoU), matching the drift-accumulation profile.

Relationship to CT-CR. CT-CR *supervises* seg pixels using det class labels; CTPV *verifies* det class labels using seg posteriors. The two operate in opposite information directions and are complementary: CT-CR propagates det evidence into the seg head, CTPV filters seg-inconsistent det evidence *before* it enters CT-CR or CT-CL. Empirically they stack: V2 + CT-CL + CT-CR alone gives 37.20 / 34.14; adding CTPV gives **37.43 / 34.42** (Table 4 headline row).

Why it’s not just detection post-processing. CTPV depends on the teacher’s segmentation posterior, which is a target-domain-adapted quantity in CTTA. In single-task detection (no seg posterior) there is no analogous signal — CTPV requires the second head. This is again the general XVA pattern: an inference-time signal that only opens up with an MTL source.

6. Full objective

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \mathcal{L}_{\text{seg}} + \lambda_{\text{CL}} \mathcal{L}_{\text{CT-CL}} + \lambda_{\text{CR}} \mathcal{L}_{\text{CT-CR}}$$

with $\lambda_{\text{CL}} = 0.5$, $\lambda_{\text{CR}} = 0.3$. STR acts on the optimiser output between backward and next forward passes; CTPV acts on the teacher’s pseudo-labels before losses are computed. Neither STR nor CTPV appears in $\mathcal{L}$.

Confidence gating (score-EM, SCORE_EM=0.5, SCORE_THRESH=1.4) determines whether $\mathcal{L}$ is backpropagated at all for the current image; on stable teacher confidence the step is skipped and STR still runs. This gating rules XVA out on frames that already look adapted, which is

the same compute-preservation trick that AMROD uses for its own loss but here also prevents CT-CL from over-tightening on easy frames.

7. Framework specialisations (deployment variants)

The framework degenerates gracefully to single-task settings. All three variants use the same CTCMT_MTL meta-architecture; the config flags CTCMT_DET_ONLY / CTCMT_SEG_ONLY gate the branches:

Variant	Task	Source	XVA components live
CTCMT_Det	detection	Mask R-CNN R-50 FPN	STR only (no second view for CT-CL/CT-CR)
CTCMT_Seg	segmentation	Semantic FPN R-50	STR + a per-class prototype anchor + trimmed 6-view aug-avg teacher
CTCMT-MTL (full)	joint det + seg	Panoptic-FPN R-50 MTL	CT-CL + CT-CR + STR + CTPV

The single-task specialisations are what one gets when the framework is applied to a source model whose architecture provides no second view. Neither variant is our primary contribution; both are **provided as sanity checks that the framework does not lose ground on single-task workloads** (CTCMT_Det beats AMROD by +0.92 AP50 on the same MR source, CTCMT_Seg beats CoTTA by +1.28 mIoU on the same Sem-FPN source).

CTCMT_Seg extras (per-class prototype anchor + trimmed multi-scale aug-avg) are single-task-only tricks that a segmentation-only adapter needs to compensate for the missing det branch. They are described briefly in Appendix A.

8. Hyperparameters

Symbol	Value	Meaning
β	0.9998	Teacher EMA decay
ρ	0.01	Base stochastic restore rate (task-head weights)
η	0.1	STR shared-trunk multiplier
τ	0.07	SupCon temperature
λ_{CL}	0.5	CT-CL weight
λ_{CR}	0.3	CT-CR weight
$\tau_{init}, \tau_{min}, \tau_{max}$	0.80 / 0.70 / 0.90	Dynamic det threshold init and bounds
α, γ	1.3 / 0.95	Dynamic threshold update coefficients
s_0^{em}	0.5	Initial teacher-score EMA
batch size	1	Standard online CTTA convention
optimizer	SGD	lr = 10^{-3} , wd = 10^{-4} , no warmup

9. Empirical isolation of XVA (Table 4 ablation, 3 seeds each)

Increment	AP50	mIoU
MTL baseline (soft-CE det + soft-CE seg, flat restore)	34.09 ± 0.05	31.79 ± 0.06
+ STR (shared-trunk restore, alone)	34.89 ± 0.07	32.41 ± 0.07
+ CT-CL (instance XVA, alone)	35.98 ± 0.12	33.23 ± 0.05
+ CT-CL + CT-CR + STR (three-component XVA)	37.20 ± 0.06	34.14 ± 0.03
+ CT-CL + CT-CR + STR + CTPV (full XVA)	37.43 ± 0.07	34.42 ± 0.06
Full delta over baseline	+3.34	+2.63
Positive synergy over additive (3-component)	+0.42	+0.29
CTPV incremental gain	+0.23	+0.28

All XVA components produce individually significant deltas ($\gg 3\sigma$ against seed variance), and the three-component combination shows a positive super-additive gain. CTPV adds a further significant boost with essentially no interaction (the +0.23 / +0.28 deltas fall inside 3σ of the tightened final variance).

10. Framework relationship to prior CTTA

Prior CTTA methods are **instances of teacher-student adaptation with a single head**. Our framework is **teacher-student adaptation with XVA on top of two heads**. When restricted to a single head, XVA vanishes and the framework recovers the corresponding prior work — up to specific plumbing choices (which teacher scheduling, which restore, which thresholding). We use this to place prior work formally:

Method	Setting	Corresponds to XVA framework with...	Difference from our best
CoTTA (Wang <i>et al.</i> 2022)	seg	seg branch only, flat restore, 14-view aug-avg	no XVA; flat restore under-corrects trunk under MTL
CT-CMT (Moraiti <i>et al.</i> 2026)	det (YOLOX)	det branch only, dyn-thresh, flat restore	no XVA (no seg head); single-task backbone
AMROD (Wei <i>et al.</i> 2026)	det	det branch only, dyn-thresh + score-EM, flat restore	no XVA (no seg head); flat restore
W3TTA-OD (Yoo <i>et al.</i> 2024)	det	det branch only, feature-stats KL, no restore	no XVA; overfits first-encountered domain

Method	Setting	Corresponds to XVA framework with...	Difference from our best
TENT (Wang <i>et al.</i> 2021)	seg	seg branch only, entropy, no teacher, no restore	no XVA, no anchor $\rightarrow$ collapse
CT-CMT-MTL (ours)	det + seg	CT-CL + CT-CR + STR + CTPV on top of the above plumbing	—

The framework is thus **an axis-extension** of prior work: same teacher- student backbone, one new axis (cross-task view alignment) that only opens up with an MTL source. This distinction is worth flagging in the intro to prevent readers from parsing our contribution as “we combine AMROD det and CoTTA seg”. We do use those pieces as pluggable teacher-student plumbing, but the framework’s value is what XVA adds on top of *any* such plumbing — we demonstrate this by verifying the ablation deltas of CT-CL, CT-CR, and STR are all individually large ($\gg 3\sigma$) even in the absence of the others (Table 4).

11. Component-attribution matrix

Every design choice has ancestors. The table below states the closest prior work for each component of our system, whether it is ours, and — for the “ours” rows — what specifically we did that is not already in the cited ancestors.

Component	Closest prior work	Ours	What is new (if any)
Mean-teacher (student + EMA teacher)	Tarvainen & Valpola 2017; used by MoCo v3, CoTTA, AMROD, TENT-teacher variants		— (unchanged)
Flat stochastic restore	CoTTA (Wang <i>et al.</i> CVPR 2022); ancestor: EWC (Kirkpatrick <i>et al.</i> 2017), SWA (Izmailov <i>et al.</i> 2018)		— (unchanged)
STR — shared-trunk restore split	EWC per-parameter importance; LwF (Li & Hoiem 2017); ULMFiT layer-wise LR (Howard & Ruder 2018); Adapters (Houlsby <i>et al.</i> 2019, different rates for shared vs task-specific components)	yes	first to apply the shared-vs-task-head distinction to CoTTA’s <i>restore rate</i> (not learning rate, not Fisher weight) in the CTTA regime

Component	Closest prior work	Ours	What is new (if any)
Dyn-threshold per-class pseudo-labels	AMROD (Wei <i>et al.</i> ESWA 2026); FreeMatch (Wang <i>et al.</i> NeurIPS 2023, self-adjusted per-class); FlexMatch (Zhang <i>et al.</i> NeurIPS 2021, curriculum); CT-CMT (Moraiti <i>et al.</i> 2026 EJA)		— (adopted verbatim from AMROD)
Score-EM confidence gate	AMROD; SAR (Niu <i>et al.</i> ICLR 2023, entropy-based sample selection); EATA (Niu <i>et al.</i> ICML 2022, uncertainty-aware TTA)		— (adopted from AMROD)
Supervised contrastive loss form	SupCon (Khosla <i>et al.</i> NeurIPS 2020); InfoNCE (van den Oord <i>et al.</i> 2018); SimCLR (Chen <i>et al.</i> 2020)		— (loss form imported unchanged)
CT-CL — det $\leftrightarrow$ seg cross-task view SupCon	Cross-Task Consistency (Zamir <i>et al.</i> CVPR 2020, image-space consistency across tasks like depth $\leftrightarrow$ normals); CMC (Tian <i>et al.</i> ECCV 2020, multi-view SSL where views are augmentations of one input); CLIP (Radford <i>et al.</i> 2021, image $\leftrightarrow$ text contrastive); MulT (Bhattacharjee <i>et al.</i> CVPR 2022, MTL transformer with cross-task attention)	yes	first CTTA-time cross-task-head contrastive; novel per-object seg view (teacher-posterior-weighted student features)

Component	Closest prior work	Ours	What is new (if any)
CT-CR — dense bbox → seg-class consistency	Cross-Task Consistency (Zamir 2020); BoxSup (Dai <i>et al.</i> ICCV 2015, box → sem-seg for source training); BoxInst (Tian <i>et al.</i> CVPR 2021, box-supervised instance seg); Panoptic training losses (Kirillov <i>et al.</i> CVPR 2019); det ↔ seg knowledge distillation (Chen <i>et al.</i> 2020)	yes	first use of bbox → seg-class CE as an <i>adaptation</i> signal (prior work uses it only at source training time)
CTPV — cross-task pseudo-label verification	Confidence-based sample selection in TTA (SAR, Niu <i>et al.</i> ICLR 2023; EATA, Niu <i>et al.</i> ICML 2022); mutual-verification in co-training (Blum & Mitchell 1998); teacher-student pseudo-label filtering (FixMatch, Sohn <i>et al.</i> NeurIPS 2020)	yes	first pseudo-label filter that uses a <i>different task head’s posterior</i> as the verification signal in CTTA; extends the co-training principle to cross-task heads of a single model
Per-class prototype anchor (CTCMT_Seg)	AdaContrast (Chen <i>et al.</i> CVPR 2022); ProtoNet (Snell <i>et al.</i> 2017); iCaRL prototype rehearsal (Rebuffi <i>et al.</i> 2017); ProDA (Zhang <i>et al.</i> CVPR 2021, prototype-based UDA-seg)	yes	CTTA-time (online, no rehearsal buffer), per-pixel, gated by teacher confidence
Trimmed 6-view aug-avg teacher (CTCMT_Seg)	CoTTA (Wang 2022, 14 views); classical TTA (Krizhevsky <i>et al.</i> 2012); Learning-to-TTA (Molchanov <i>et al.</i> 2020, learned aug selection)	yes	empirical trim decision (CoTTA’s aggressive aug-avg hurts night; 6 views retain fog gain and clean night degradation)

Component	Closest prior work	Ours	What is new (if any)
MTL adaptation with a single shared model (CTTA)	Original MTL (Caruana 1997); MTAN (Liu <i>et al.</i> CVPR 2019, task attention on shared features); UberNet (Kokkinos 2017); MulT (2022). CTTA angle: Continual MTL (Sun <i>et al.</i> 2020) trains under continual task addition; VDP (Marsden <i>et al.</i> CVPR 2024, MTL TTA but only vision-language)	yes	first CTTA (continual domain shift, target-only, no source data) for a shared det + seg model on ACDC-scale weather benchmarks
Panoptic-FPN backbone	Kirillov <i>et al.</i> CVPR 2019		— (source model architecture)
Continual eval protocol (ACDC fog → night → rain → snow)	CoTTA / AMROD / W3TTA-OD		— (adopted for direct comparability)

Reading the matrix. Every “yes” row has genuine ancestors — we do not claim any of the underlying primitives (contrastive learning, prototype anchors, per-parameter restore, cross-task consistency) as our invention. What is new is the specific instantiation and application context: applying these primitives to (a) the CTTA regime, (b) *at test time*, (c) *with the XVA construction of two views per object*, in a domain where prior CTTA work has only ever used one head’s output.

12. Fault-mode analysis of prior CTTA

We now formalise the two failure modes announced in §1 by pointing to the exact code sites in the released implementations, giving each a numerical signature from our ACDC results, and stating which XVA primitive fixes it.

12.1 CoTTA vulnerabilities ([cotta_semseg.py](#))

CoTTA-1. Aug-averaging triggered by the source anchor’s confidence. Line 258 (`_anchor_confidence`) + 269: `low_mask = conf < self.conf_threshold`, where `conf` is the frozen source’s own softmax max-prob. On rain and snow the source is confidently *wrong* (high softmax, wrong class); anchor confidence stays above threshold and aug-avg **never fires** on the images where it would help most. *Numerical signature.* CoTTA fog 46.24 → snow 35.79 (drop of 10.45). *XVA fix.* CTCMT_Seg triggers aug-avg on **teacher** entropy, not anchor confidence; snow +3.03 mIoU (Table 2).

CoTTA-2. 14-view aug-avg forward budget. Lines 216–224. Three scales × 5 flips/aug ops → 14 extra teacher forwards on every triggered image, ~1.4 GB extra activations at 1024×2048. *Numerical signature.* CoTTA’s night 22.24 shows aug-avg firing on many hard images yet giving no visible gain over source-only 22.60. *XVA fix.* CTCMT_Seg trims to 6

views (3 scales $\times$ 2 flips). Fog gain retained ($46.38 \geq 46.24$), night cleaner ($22.81 > 22.24$), less compute.

CoTTA-3. Uniform stochastic restore. Line 158: `mask = (torch.rand_like(p) < rst).float()` — flat rate $\text{rst} = 0.01$ over every trainable weight. Shared backbone and task-head weights treated identically, even though the shared backbone is on the gradient path of every loss term. *Numerical signature.* CoTTA drift grows visibly across weathers: rain +0.61 vs source, snow -1.03 vs source (the restore fails to protect against accumulated backbone drift). XVA fix. **STR** — split the restore into rst for heads and $\text{rst} \times \eta$ ($\eta = 0.1$) for backbone/FPN so the shared trunk gets restored $10\times$ more often per weight. Ablation: +0.80 AP50 / +0.62 mIoU standalone (Table 4 row 2).

CoTTA-4. Every pixel weighted equally in soft-CE. Line 279: `loss = -(teacher_probs.detach() * log_probs).sum(dim=1).mean()`. Pixels where the teacher is genuinely uncertain (fog haze, boundary regions) inject noise proportional to their pixel count. *Numerical signature.* Uniform per-pixel loss propagates fog-boundary drift into later weathers. XVA fix. **CT-CR** confines dense supervision to pixels *inside teacher-detected boxes*, a natural high-confidence filter derived from the parallel head.

CoTTA-5. Only one supervision signal. Every loss term is teacher-student on one head. When the teacher goes wrong in one direction, the student follows. XVA fix. **CT-CL + CT-CR** provide orthogonal signals via the second head. Cross-task disagreement fires *before* either head’s own confidence drops.

12.2 AMROD vulnerabilities ([amrod.py](#))

AMROD-1. Strong-augmentation asymmetry backfires on hard weather. Line 519: `images = self.model.preprocess_image(batched_inputs, strong_aug=True)` — student sees color jitter / cutout / noise; teacher sees clean input. On rain and snow the “clean” input already contains augmentation-like corruption; layering synthetic augmentation on top pushes student features out of the teacher’s clean-prediction manifold. *Numerical signature.* AMROD night 15.23 (barely above source-only 13.45), consistent with strong-aug corrupting already-corrupted dark low-contrast images. XVA fix. CT-CMT-MTL feeds the same image to both branches and gets consistency from the cross-task view instead of augmentation. Night 17.18 vs 15.23 (+1.95).

AMROD-2. One-step Fisher restore is very noisy. Lines 543–549 store `p.grad.data.clone().pow(2)` from a *single* backward pass as an EWC-style importance estimate. Then line 573 uses `find_weight_quantile` to keep the top-quantile weights. The same parameter can be “important” one step and “restored” the next because per-step gradient variance is high. *Numerical signature.* AMROD 3-seed std: fog 0.23, night 0.19 vs CT-CMT-MTL’s 0.10, 0.11 — twice the seed noise on the same task. XVA fix. **STR** is structural (shared vs task-head split) instead of statistical (Fisher). One deterministic hyperparameter, no gradient estimation, seed-stable.

AMROD-3. Symmetric score-EM gate stops adaptation at the shift boundary. Line 500: `if (mean_all / self.score_em) > self.score_thresh or (self.score_em / mean_all) > self.score_thresh: ... return outputs`. The gate skips whenever the current score deviates in **either direction** from the running EMA. But domain shift is **asymmetric**: teacher scores drop suddenly at fog→night, then recover. The gate spams “skip” during that recovery window — exactly when the model needs to adapt. *Numerical signature.* AMROD night 15.23 vs W3TTA-OD 18.33 on the same domain (W3TTA-OD has no such gate). XVA fix. CT-CL and CT-CR fire on the *segmentation* branch, which does not consult the det gate. Cross-task signal keeps the shared backbone adapting even when det pseudo-labels vanish.

AMROD-4. The “contrastive” loss is a distillation echo. Line 534: `loss_dict["AMROD"] = self.loss_weight * self.AMROD_contrastive_loss(s_query, t_query)`. `s_query` and `t_query` are the *same* teacher-proposed box features projected through `self.query_head`,

once by student and once by teacher. There is no positive/negative diversity beyond teacher’s own emission — this is a similarity-preserving distillation, not a genuine contrastive. *XVA fix*. **CT-CL** builds two genuinely independent views (det-RoI vs seg-mask-pooled) from **different heads** of the same student. Multi-view contrastive rather than distillation.

AMROD-5. Narrow dynamic threshold band. Lines 46, 60, 61: THRESHOLD_INIT = 0.80, THRESHOLD_MIN = 0.70, THRESHOLD_MAX = 0.90. On night the teacher scores collapse to the 0.4–0.7 band; the threshold floor at 0.70 rejects almost every pseudo-box, and the detection branch stops receiving any supervision. *Numerical signature*. AMROD night 15.23, the largest per-domain gap to any competitor. *XVA fix*. CT-CR supplies dense pixel supervision that survives even when det pseudo-boxes vanish — as long as *some* box is kept, all its pixels reach the segmentation head.

AMROD-6. Teacher misses accepted as ground truth AND teacher misclassifications accepted as ground truth. Lines 517–522: $t_proposals$ from the teacher become RPN + RoI GT for the student. Objects the teacher missed at high threshold produce **zero gradient** (misses); objects the teacher mis-classifies produce **wrong-class gradient** (mis-labels). The student can never learn to detect what the teacher does not, and it actively learns to mis-classify what the teacher mis-classifies. *XVA fix (two-sided)*. **CT-CR** enforces bbox-interior seg consistency, so when the seg head sees an object that det missed, the shared trunk still gets supervision — the object leaks back into the next step’s det features. **CTPV** consults the seg posterior over the box interior and *rejects* the box if the seg head disagrees about the class, cutting off the wrong-class-gradient channel before it enters CT-CL / CT-CR. Empirically CTPV alone contributes **+0.23 AP50 / +0.28 mIoU**, with the largest per-domain gains on snow (+0.65 AP50 / +0.84 mIoU), matching the drift-accumulation profile.

AMROD-7. No recovery from teacher collapse. After the EMA teacher drifts far under strong-aug on later domains, the 0.9998 EMA and 0.01 restore are both too slow to pull it back. *XVA fix*. **STR** with $\eta = 0.1$ acts as an implicit re-anchor: the shared trunk is restored to source $\sim 10\times$ more often than the heads, resetting drift while heads keep their domain-specific tuning. CT-CMT-MTL snow 43.40 vs AMROD 40.99 (+2.41).

12.3 Consolidated fault → fix table

Failure mode	Prior weakness	XVA fix	Ablation evidence
FM-1 teacher-collapse	CoTTA-1 (anchor-triggered aug-avg silent on hard weather)	CTCMT_Seg trigger on teacher entropy	+3.03 mIoU on snow (Tab 2)
FM-1 teacher-collapse	CoTTA-2 (14-view budget)	CTCMT_Seg 6-view trim	3× cheaper, $\geq$ fog, +0.57 night
FM-1 teacher-collapse	CoTTA-4 (pixel-uniform soft-CE)	CT-CR dense box-interior CE	+1.44 mIoU standalone (Tab 4)
FM-1 teacher-collapse	CoTTA-5 / AMROD-3 (single supervision signal)	CT-CL + CT-CR orthogonal views	+1.89 AP50 / +1.44 mIoU standalone (Tab 4)
FM-1 teacher-collapse	AMROD-1 (strong-aug on hard weather)	Clean-clean pair, cross-task compensates	+1.95 AP50 night (Tab 1)
FM-1 teacher-collapse	AMROD-4 (pseudo-contrastive is distillation)	Real 2-view SupCon	(subsumed in CT-CL delta)

Failure mode	Prior weakness	XVA fix	Ablation evidence
FM-1 teacher-collapse	AMROD-5 (narrow threshold floor)	CT-CR dense supervision survives det collapse	(subsumed in CT-CR delta)
FM-1 teacher-collapse	AMROD-6 (teacher misses + mis-labels flow through)	CTPV vetoes seg-disagreeing det boxes	+0.23 AP50 / +0.28 mIoU standalone; +0.65 / +0.84 on snow (Tab 4)
FM-2 drift under uniform restore	CoTTA-3 (flat restore)	STR shared $\times \eta$ split	+0.80 AP50 / +0.62 mIoU standalone (Tab 4)
FM-2 drift under uniform restore	AMROD-2 (per-step Fisher noise)	STR deterministic structural split	seed std halved (Tab 3)
FM-2 drift under uniform restore	AMROD-7 (no re-anchor)	STR $\times \eta$ acts as implicit re-anchor	+2.41 AP50 snow (Tab 1)

Two failure modes $\rightarrow$ four primitives (CT-CL, CT-CR, STR, CTPV) that together address every row.

13. Anticipated reviewer objections

“You still borrow the teacher, the restore, and the dyn-threshold from prior work.” True — and stated. Our claim is not novelty of the teacher-student mechanism but novelty of the XVA supervision it enables in MTL. To verify this claim we re-implement the borrowed plumbing exactly and show that removing XVA (Table 4 baseline row) reproduces AMROD-class det results and CoTTA-class seg results on the MTL source.

“CT-CR looks like a special case of pseudo-instance segmentation.” CT-CR does not attempt to predict masks; it enforces that the seg head’s existing per-pixel output be spatially aligned with detection geometry. It is a cross-task consistency loss, not a mask predictor.

“STR is a hyperparameter tweak.” STR is available *only* because MTL distinguishes shared trunk from task heads. The corresponding hyperparameter tweak in a single-task model is a global ρ scan, which we ran and found no improvement over CoTTA’s $\rho = 0.01$ — see the “flat restore baseline” row of Table 4.

“CT-CL is just SupCon.” SupCon is the loss form; XVA is the view construction. In particular, z_{seg} is a novel per-object feature that uses the teacher’s segmentation posterior as an attention mask over student features. That construction — and its use as the second view for per-object SupCon — is new.

14. Implementation

- Meta-architecture: [detectron2/detectron2/modeling/meta_arch/ctcmt_mtl.py](#) (751 lines)
- Single-task seg variant: [detectron2/detectron2/modeling/meta_arch/ctcmt_seg.py](#) (309 lines)
- Config knob `MODEL.CTCMT_STUDENT_META_ARCH` selects PanopticFPN (default, MTL) or GeneralizedRCNN (single-task det, CTCMT_Det), so the same meta-arch handles all three deployment modes with only YAML changes.
- In the code, **STR** is `CTCMT_CROSS_TASK_FISHER=True` with `CTCMT_BACKBONE_RST_FACTOR=0.1`; **CT-CL** is `CTCMT_CTCL_ENABLED=True` with `CTCMT_CTCL_SEG_VIEW=True`; **CT-CR** is `CTCMT_WEIGHT_CTCR=0.3`.